

American Option Pricing via Least-Squares Monte Carlo using ML-based Continuation Regression

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<https://youtu.be/ESI8ocXSk0s>

Problem & Challenges

European Options

- Exercise only at maturity
- Closed-form solution (Black-Scholes)
- Straightforward discounting

American Options

- Exercise at any time before maturity
- No closed-form solution
- Optimal stopping problem

The Core Challenge

At each time step, the holder must decide: exercise now or continue?

This requires estimating $E[\text{future payoff} \mid \text{current state}]$ — the continuation value.

Accurate estimation of this conditional expectation is what drives pricing quality.

Motivation

ML Relevance

- LSMC's core step is a regression problem
- Supervised learning: predict continuation value from state
- When does ML help vs. when is it pointless?

Industry Relevance

- LSMC is the industry standard for complex derivatives
- Scales to multi-factor, path-dependent settings
- American-style early exercise is ubiquitous

Research Goal

- Reproduce Longstaff & Schwartz (2001) baseline
- Replace OLS with Ridge, Lasso, Random Forest, Gradient Boosting
- Evaluate price accuracy, convergence, runtime, and sensitivity

Related Work

1

Longstaff & Schwartz (2001)

Valuing American Options by Simulation: A Simple Least-Squares Approach

The paper being reproduced. Introduced LSMC — polynomial regression to estimate continuation values from simulated paths.

2

Andersen & Broadie (2004)

Primal-Dual Simulation Algorithm for Pricing Multidimensional American Options

Produces lower and upper bounds via primal-dual ideas, providing confidence intervals for the true option value.

3

Haugh & Kogan (2004)

Pricing American Options: A Duality Approach

Dual martingale formulation yielding upper bounds. A principled benchmark against primal simulation methods like LSMC.

The LSMC Method

1

Simulate

Generate N paths of stock price S via Geometric Brownian Motion

2

Backward Induction

Start near maturity, step backward through exercise dates

3

Regression

For in-the-money paths, regress discounted future payoffs on basis functions of S to estimate $E[V|S]$

4

Exercise Decision

Exercise if immediate payoff $>$ estimated continuation value; otherwise continue

5

Price

Average discounted realized cashflows across all paths

Key Insight: Step 3 is the regression — replace OLS polynomial with ML regressors

ML Extension: 5 Regression Methods

Linear

OLS (deg 2)

Baseline polynomial basis: $\{1, S, S^2\}$

Linear

Ridge

L2 regularization — shrinks coefficients toward zero

Linear

Lasso

L1 regularization — can zero out coefficients

Tree

Random Forest

Ensemble of decision trees, bagging for variance reduction

Tree

Gradient Boosting

Sequential trees, each correcting previous residuals

Experimental Setup

Baseline Parameters (Longstaff & Schwartz Table 1)

Parameter	Value
Initial stock price (S_0)	36
Strike price (K)	40
Risk-free rate (r)	0.06
Volatility (σ)	0.20
Time to maturity (T)	1 year
Exercise dates	50
L&S benchmark price	4.478

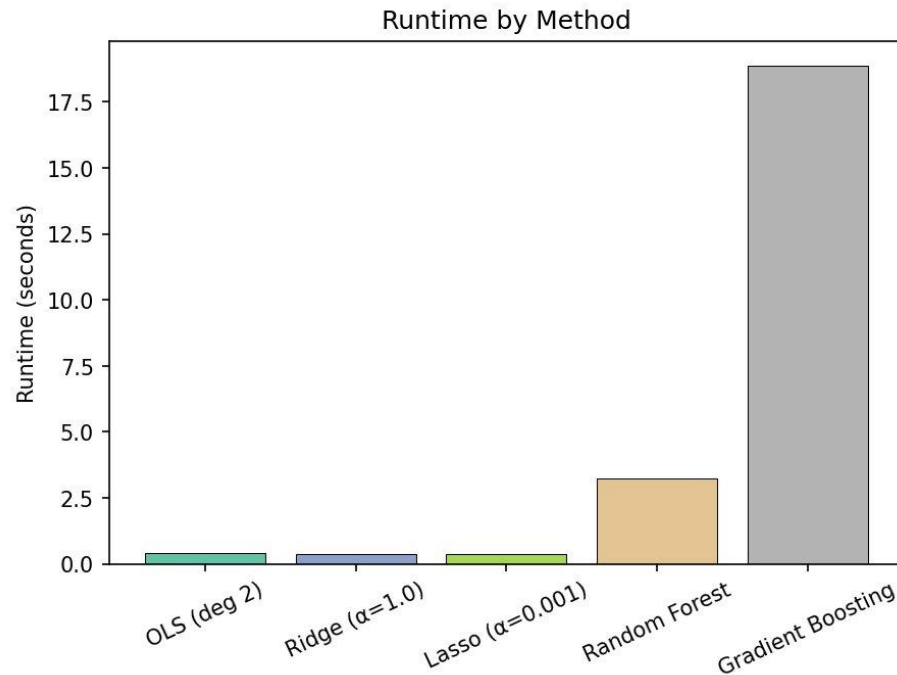
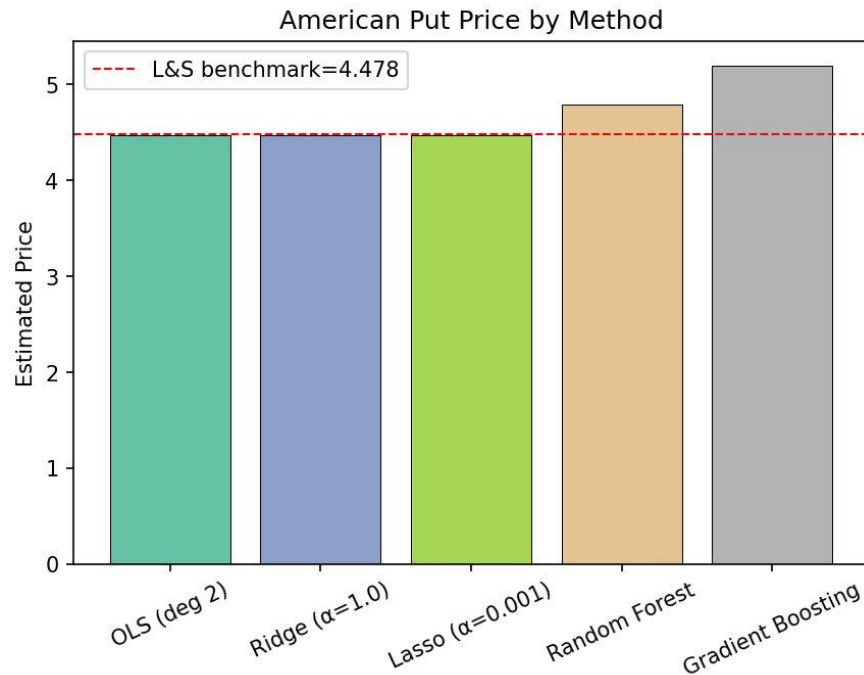
Five Experiments

- Benchmark comparison — accuracy & runtime at baseline
- Convergence — price vs number of simulated paths
- Hyperparameter sensitivity — Ridge α , Lasso α , RF depth, GB learning rate, etc.
- Accuracy vs speed tradeoff — Pareto frontier
- Option parameter robustness — 6 scenarios $\times$ 5 methods

Note: Linear methods run on 100K paths. Tree methods capped at 10K due to runtime constraints (~19s for GB at 10K).

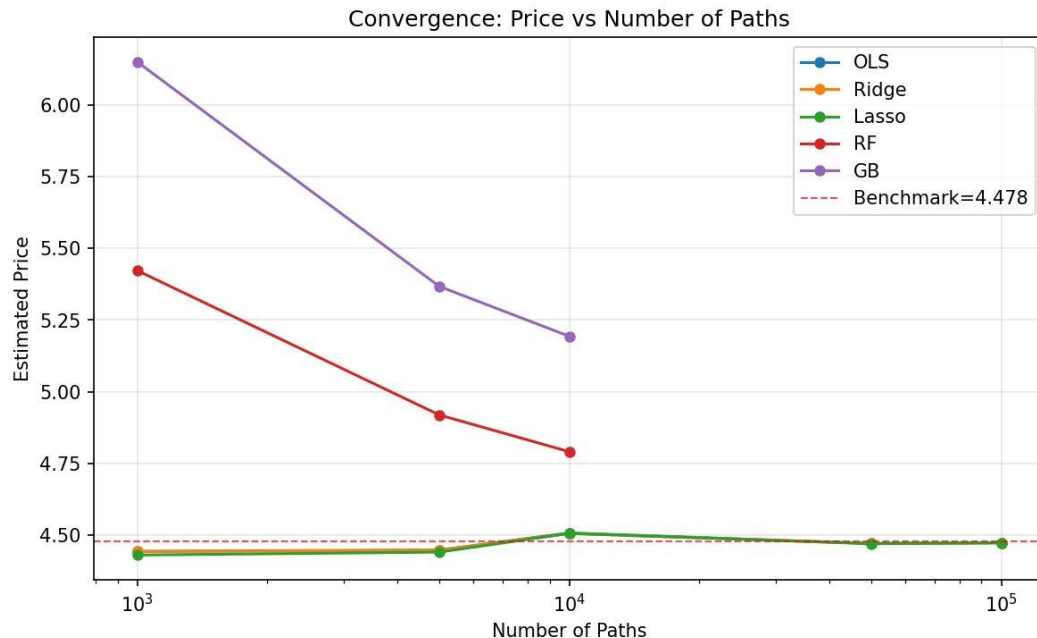
Results: Benchmark Comparison

LSMC Regression Methods — Benchmark Comparison



Linear methods: < 0.11% error, < 0.45s | RF: +7.0% error, 3.2s | GB: +16.0% error, 18.9s

Results: Convergence



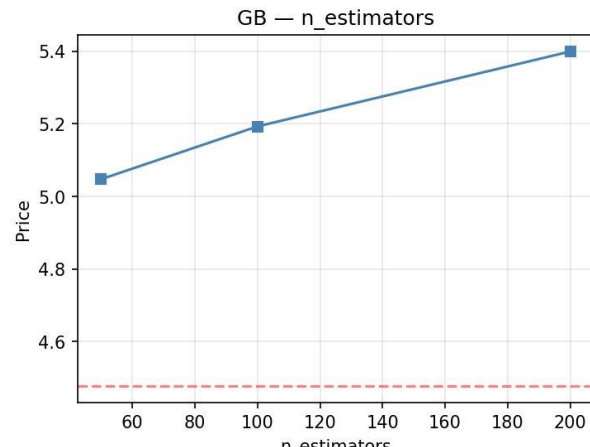
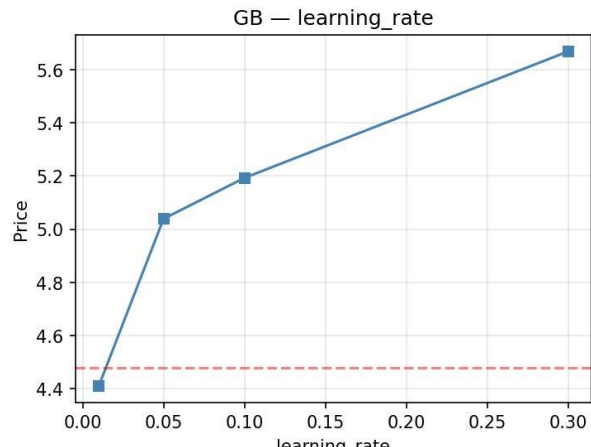
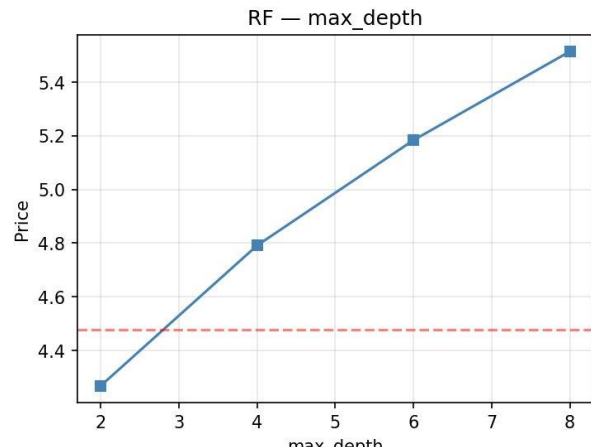
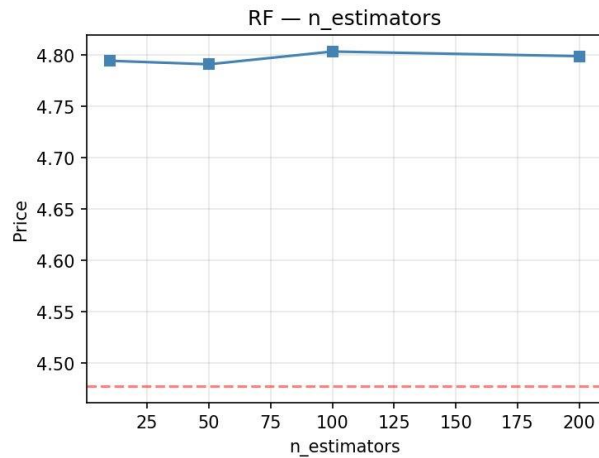
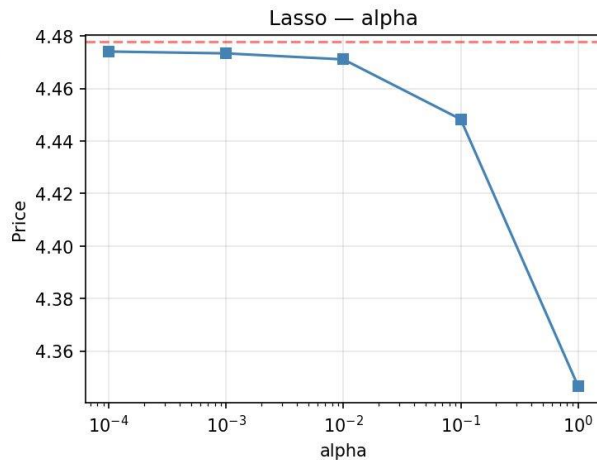
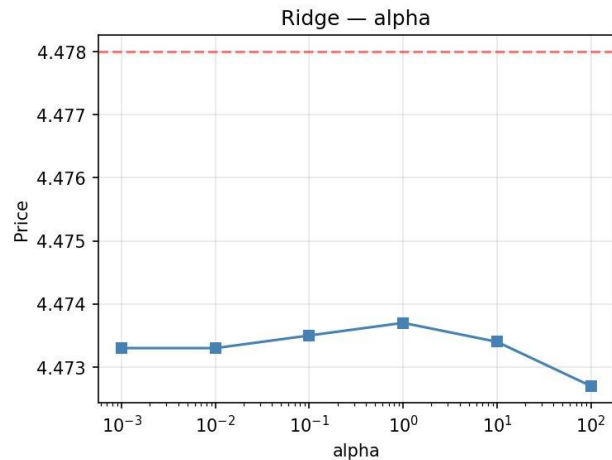
Key Findings

- OLS/Ridge/Lasso converge by ~1K paths — polynomial basis is very sample-efficient
- RF: starts at 5.42 (1K), slowly drops to 4.79 (10K)
- GB: starts at 6.15 (1K), still at 5.19 (10K) — overfits small samples badly
- Tree methods need far more paths, but polynomial bases match the smooth 1D continuation surface

The smooth continuation value $E[V|S]$ for a standard put is well-approximated by a low-degree polynomial — tree methods waste capacity.

Results: Hyperparameter Sensitivity

Hyperparameter Sensitivity



Hyperparameter Insights

Linear Methods

Ridge α : Almost no effect — price varies by ~ 0.001 across 5 orders of magnitude. The problem is too well-behaved for regularization to matter.

Lasso α : Stable at low α , but at $\alpha=1.0$, price drops to 4.35 — Lasso zeros out polynomial terms too aggressively.

Tree Methods

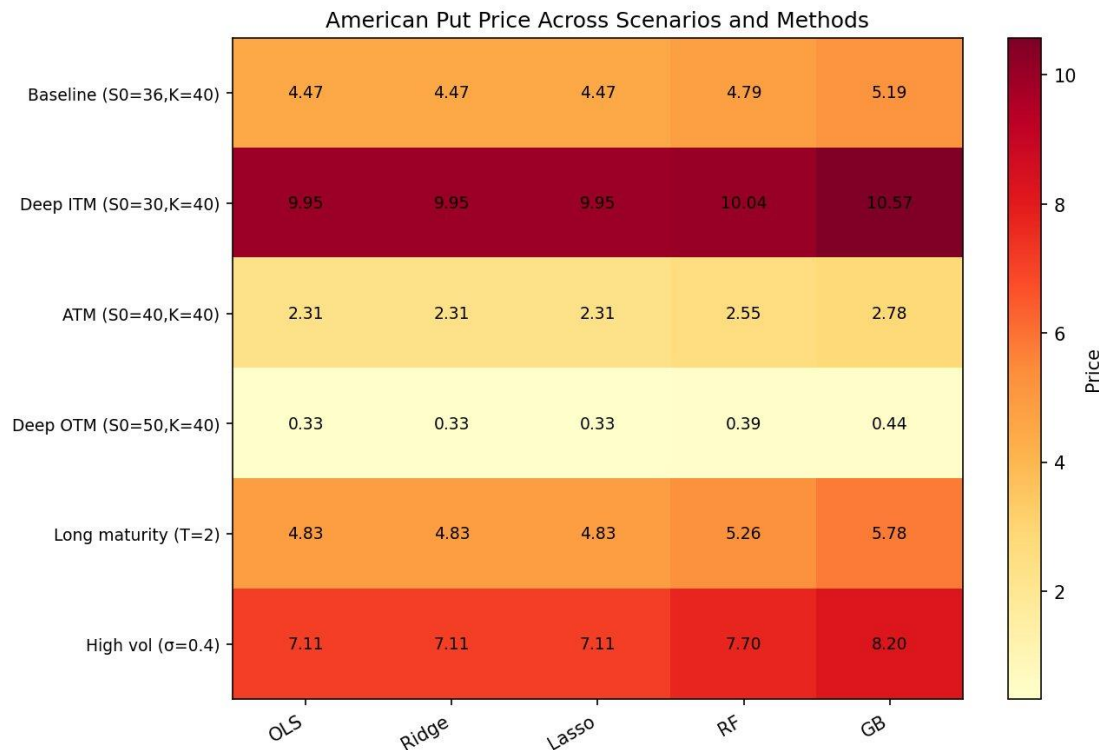
RF `max_depth`: Critical — depth 2 underfits (4.27), depth 8 overfits (5.51). Sweet spot $\sim 3-4$.

GB `learning_rate`: Most sensitive. `lr=0.01` gives 4.41 (near benchmark!), `lr=0.3` gives 5.67.

GB `n_estimators`: More rounds = worse (5.05 $\rightarrow$ 5.40). Combined with high `lr`, more rounds = more overfitting.

GB with `lr=0.01` proves tree methods CAN work — default hyperparams are the problem, not the model class.

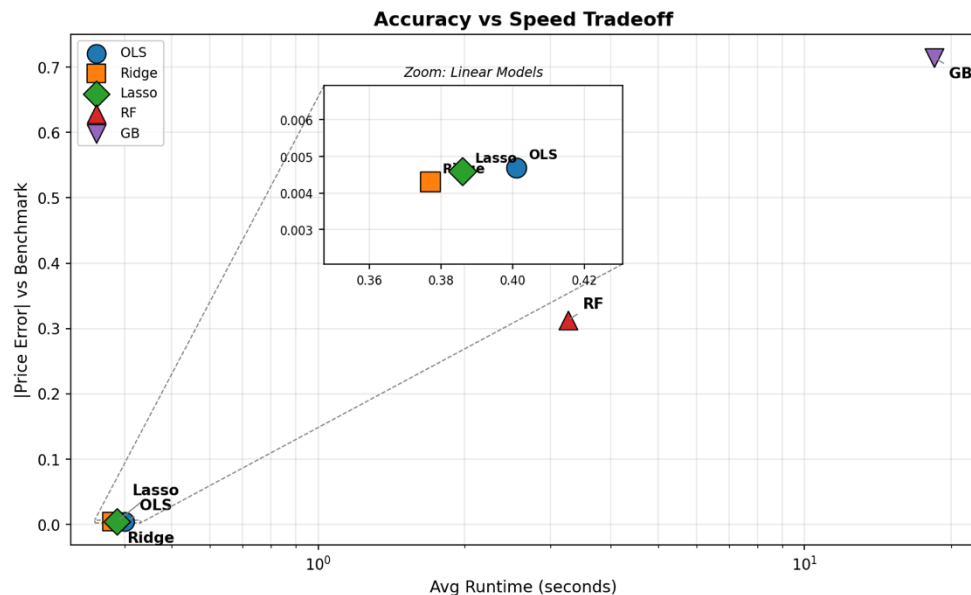
Results: Option Parameter Robustness



Observations

- OLS/Ridge/Lasso always agree — virtually identical prices across all 6 scenarios
- RF and GB systematically overprice in every scenario
- Overpricing gap grows with complexity: +0.72 baseline $\rightarrow$ +1.09 high vol
- Deep ITM puts most expensive (~9.95–10.57), consistent with theory
- High vol increases price (~7.1 vs 4.47 baseline)

Results: Accuracy vs Speed Tradeoff



Pareto Analysis

- Bottom-left = ideal (fast + accurate)
- OLS/Ridge/Lasso cluster in the ideal zone — fast and accurate, practically interchangeable
- RF: moderate error, moderate speed
- GB: slowest and least accurate at default settings
- For standard 1D LSMC, linear methods fully dominate the Pareto frontier

Tree methods would need a non-linear, multi-dimensional state space to justify their cost.

Key Observations

1

Polynomial regression is optimal for 1D American puts

OLS, Ridge, and Lasso all match the L&S benchmark to within 0.11%. The smooth continuation surface is perfectly captured by $\{1, S, S^2\}$.

2

Tree methods systematically overprice

Continuation value overfitting triggers premature exercise, inflating prices by 7–16%. The bias is consistent across all tested scenarios.

3

Convergence speed differs dramatically

Linear methods converge at ~1K paths. Trees need 10K+ and still haven't fully converged — a fundamental bias-variance tradeoff.

4

Hyperparameter tuning can rescue tree methods

GB with $lr=0.01$ achieves ~1.5% error, proving the model class isn't inherently broken — but defaults fail badly.

5

Runtime gap is 50×

Linear methods run in <0.45s. GB takes ~19s at only 10K paths. For production use, the cost is hard to justify in 1D.

What I Built vs What I Used

Existing Work (Reproduced)

- Longstaff & Schwartz (2001) LSMC algorithm and benchmark values (Table 1)
- GBM simulation framework for stock price paths
- Polynomial basis (OLS deg 2) for continuation value estimation
- Scikit-learn implementations of Ridge, Lasso, Random Forest, Gradient Boosting

My Contribution (Original)

- End-to-end Python implementation of the full LSMC pipeline from scratch
- Integration of 4 ML regressors into the backward induction loop
- 5-experiment evaluation framework: benchmark, convergence, hyperparameter sweeps, speed-accuracy, and multi-scenario robustness
- All analysis, plots, and interpretation of why tree methods fail in 1D LSMC

Conclusion & Future Work

Conclusion

For standard 1D American put pricing via LSMC, polynomial regression (OLS, Ridge, Lasso) is the clear winner — fast, accurate, and insensitive to hyperparameters. Tree-based methods are slower, systematically biased upward, and require careful tuning. However, the ability of ML regressors to handle non-linear, high-dimensional state spaces motivates their use in more complex settings.

Future Work

- **Multi-asset options** — where polynomial bases become inadequate and trees may shine
- **Neural network regressors** — deep learning for continuation value estimation
- **Path-dependent features** — Asian options, lookbacks, where state space is richer
- **Variance reduction** — antithetic variates, control variates for tighter price estimates
- **Primal-dual bounds** — confidence intervals per Andersen-Broadie and Haugh-Kogan

Thank You

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